OpenCart
Test Plan

Manual Testing | UI Automation | API | Database

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1. Objective

This project evaluates the quality, functionality, reliability, and data integrity of the OpenCart e-commerce application through manual testing, UI automation, API testing, and database testing. It also aims to identify, document, and report defects and confirm that the application behaves as expected.

2. Scope

- Functional testing of the OpenCart storefront and Admin Panel.
- Manual testing of expected and unexpected inputs, page behavior, and user workflows.
- UI automation using Selenium WebDriver.
- API testing using Postman.
- Database testing with SQL queries to verify data accuracy, integrity, and consistency.
- Defect reporting and tracking, followed by retesting and regression testing.
- Verification of UI, API, and database interactions where applicable.
- Performance, load, and stress testing in the local test environment.
- Compatibility testing across supported browsers and devices.

3. Out of Scope

- Security penetration testing.
- Testing external payment gateways or third-party services in live environments.
- Production deployment and validation of production data.

4. Roles and Responsibilities

Role	Responsibilities
Test Lead	Define the test approach, coordinate activities, assign tasks, track progress, review defects, and prepare the final summary.
Functional Tester	Perform manual testing, design and execute test cases, validate database and API results, and report defects.
Security Tester	Test registration, login, logout, password recovery, and sessions; assess application security, identify potential vulnerabilities, and document findings.
Compatibility Tester	Verify application behavior across supported browsers, operating systems, screen sizes, and devices.

Team rules: Use only approved test accounts and environments. Do not enter real customer information, and keep passwords private. Record the browser, device, test steps, and result. Obtain written approval before conducting security or load testing on a live website.

5. Entry Criteria

- The OpenCart application and local test environment are available.
- The storefront and Admin Panel are accessible for testing.
- The database is accessible through DBeaver or an equivalent database client.
- Required testing tools are installed and configured.
- The test scope and scenarios are documented.
- Suitable test data and test accounts are prepared.

6. Exit Criteria

- All planned test cases have been executed, or any remaining cases have documented reasons.
- Test execution results have been recorded.
- All identified defects have been documented and tracked in Jira.
- Critical and high-priority defects have been resolved or formally recorded as outstanding.
- Failed test cases have been retested where applicable.
- UI automation and API test results have been documented.
- Database validation has been completed and the results recorded.
- A final test summary has been prepared, including open risks and known issues.

7. Modules to Test

Area	Checks
Storefront	Homepage, product categories and details, search, filters, cart, checkout, customer registration, login, and account management.
Admin Panel	Authentication, dashboard, catalog and product management, categories, customers, orders, and related settings.
API	Available endpoints for products, customers, authentication, carts, and orders, subject to access.
Database	Relevant tables and relationships for products, categories, customers, orders, and related records.

8. Test Environment

Item	Requirement
Application	OpenCart running locally in Docker.
Operating system	Windows.
Database	MySQL/MariaDB running in Docker.
Database client	DBeaver.
Manual testing and defect tracking	Jira.
Test case documentation	Microsoft Excel.
UI automation	Selenium WebDriver.
API testing	Postman.
Browsers	Google Chrome, Mozilla Firefox, and Microsoft Edge.
Test data	Dedicated test accounts and sample products, customers, and orders.
Environment type	Local test environment; no production data will be used.

9. Test Strategy and Types

Type	Purpose
Functional	Confirm that storefront and Admin Panel features work as expected.
Negative and boundary	Check how the application handles missing, invalid, unusually long, and unexpected input.
Security	Assess application security and document findings; penetration testing is out of scope.
Compatibility	Check the application on supported browsers, operating systems, screen sizes, and devices.
Performance, load, and stress	Assess application response under normal and increased demand in the local test environment.
API and database	Check API behavior and verify database accuracy, integrity, and consistency.
Regression	Repeat key tests after fixes or application changes.

10. Deliverables

- Approved test plan and test scenarios.
- Test data and test execution results.
- Defect reports with clear reproduction steps and supporting evidence where useful.
- Retest and regression test results.
- Final test summary covering test coverage, open issues, limitations, and remaining risks.

11. Risks and Mitigation Plan

Risk	What could happen	How to reduce it
External service unavailable	A linked service may be unavailable even when the OpenCart application is working.	Record the issue and check the external service separately.
Test data changes	Products, customers, orders, or stock may change during testing.	Use dedicated test data and record the data used.
Unclear expected behavior	Testers may interpret expected results differently.	Agree on expected results before testing begins.
Limited browser/device coverage	Some users may experience compatibility or layout issues.	Test a broad range of supported browsers and devices.
Local environment differences	The local setup may differ from the deployed application.	Record versions and settings, and confirm key behavior against agreed requirements.